Convolutional Two-Stream Generative Adversarial Network-Based Hyperspectral Feature Extraction

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***Abstract*— Hyperspectral image processing is faced with difficulties considering its redundant features and complex information. Studies on hyperspectral feature extraction in the deep learning domain have become increasingly popular. The mainstream techniques fully consider the spatial information in local neighborhoods when extracting spectral features by constructing deep neural networks. Deep generative models simulate the intrinsic structure of samples by adequately training, showing their potential values for signal processing. In this article, a convolutional two-stream network (cs^{2}GAN-FE) based on the improved Wasserstein generative adversarial network (WGAN) is proposed for unsupervised hyperspectral spatial–spectral feature extraction. The improved WGAN is composed of one generator and one discriminator; the former perceives real data distributions, and the latter determines the attribution of generated data. The designed two-stream strategy is not a simple extension of a one-stream strategy and considers both the static spectral–spatial information and the dynamic spectral reflectance variation in multiple bands. Intrinsic spatial–spectral features are extracted by the trained discriminator considering sample distributions and feature relationships. The loss function is also improved for the unique structure of cs^{2}GAN-FE. Various state-of-the-art techniques are chosen for comparison. Experimental results show the feasibility and potential of this network. Besides, experiments with the random split and the disjointed split both show that the proposed method can outperform other comparison techniques.**

***Index Terms*— Feature extraction, generative model, hyperspectral image, spectral reflectance variance, unsupervised learning.**

I. Introduction

HYPERSPECTRAL imaging technologies occupy crucial positions in the geoscience field, providing abundant research directions [1], [2]. Compared with classical RGB images, hyperspectral images comprise more discriminant information delivering complex topographic features; however, this is at the expense of hundreds of spectral redundant bands [2], [3]. As an effective solution to these problems, feature extraction maps original spectral features into specific latent spaces by transformation and then generates latent embeddings with the higher discriminant ability and less storage space [4], [5].

Recently, constructing deep neural networks (DNNs) occupies a significant position [3], [6]. As typical nonlinear models, DNNs are useful tools for hyperspectral image processing because of the complex imaging principle [7]. Compared with shallow NNs, deep models fully extract high-level features through nonlinear transformation, retaining the latent embedding and making up for the shortcomings of classical feature extraction techniques. Multiple deep learning platforms (such as HyperLabelMe [8]) are presented for feature learning to master the latent characteristics [9]. Autoencoders (AEs) and their variants [10] work well in signal compression since they reduce the redundancy under the premise of lossless data reconstruction. Convolutional neural networks (CNNs) also occupy important positions considering their superiority in spatial feature construction [11], [12].

Researchers have proposed various deep learning platforms for hyperspectral data specifically to learn the unique characteristics [13], [14]. Some AE-based variants were designed for remote sensing analysis, and they reflect the advantages of combining extraction and reconstruction. Zabalza *et al.* [15] confronted a single sample into small segments to reduce the computational complexity and provided a segmented stacked AE (S-SAE) network. Ahmad *et al.* [16] proposed an unsupervised nonlinear stacked denoising AE (DAE) with similar structures. However, these improvements cannot consider the correlation between local samples and the spatial arrangement. The spatial arrangement corresponds to the abundant spatial information in hyperspectral images, which should be fully exploited [6], and more attentions are focused on multidimensional (multi-D) convolution kernel-based studies [17]. The unique structures of CNNs make inputs with 2/3-D shape possible, and the high-level features are extracted by adjusting the shapes of inputs [18]. Zhao *et al.* [19] proposed superpixel-based multiple local CNNs for multispectral classification. Pixels were replaced by superpixels for basic analysis, and this model could extract an efficient joint latent embedding. Ma *et al.* [20] presented a double-branch multiattention (DBMA) mechanism network for hyperspectral classification to reduce the interference between different kinds of features. This method successfully extracts multiple features using two branches for classification.

At present, hyperspectral feature extraction utilizing generative models is still at the primary stage; however, it shows

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significant potential. Generative models are generally introduced for hyperspectral analysis, and some related works have been finished. Zhang *et al.* [7] directly used GAN to extract high-level features and obtained good results. This method (uWGAN for short) is the first attempt to use the Wasserstein generative adversarial network (WGAN) for hyperspectral classification. He *et al.* [21] proposed a 3-D bilateral filter-based classification technique inspired by GAN to extract the spectral–spatial features. After that, large amounts of unlabeled samples were used to train the GAN model, and labeled samples were adopted to fine-tune the trained network by adding a soft-max classifier. Zhang *et al.* [22] presented an efficient hyperspectral image retrieval method based on deep convolutional GAN (DCGAN), and extracted embeddings were mapped by t-SNE-based nonlinear manifold hashing. These single stream generative models perform well; however, using various streams to extract multiple features increasingly shows superiority [23]. Our motivation is to fully extract multiple features from different angles through generative models and combine them effectively.

As a classical technique in the behavior recognition field, the two-stream architecture (called two-stream ConvNet) works well on video classification specifically [24]. In this network, one stream is called a spatial stream, and its input is an RGB image (single frame); the other stream is designed for multiframe optical flow describing the movement of pixels. Inspired by it, we adapted this definition for the attribute of hyperspectral data and proposed a novel unsupervised convolutional two-stream WGAN (cs^{2}GAN-FE). cs^{2}GAN-FE mainly consists of two streams: static stream and dynamic stream. The former corresponds to the spatial stream in the classical two-stream ConvNet, delivering features in a single band. Since the positions of hyperspectral pixels are fixed in different bands, we defined a strategy to simulate the spectral reflectance variations of pixels within adjacent bands and the dynamic stream extracts features from these variations (rather than unprocessed adjacent bands). It is clear that our work goes beyond a simple extension from one-stream architecture to a two-stream architecture. cs^{2}GAN-FE comprises one generator and one discriminator (both have two streams). Considering the adversarial strategy, the former aims to generate deceptive samples for the latter. The proposed loss function ensures the Lipschitz smoothness of the discriminator, and the trained discriminator is transformed into our feature extractor. The implementation of cs^{2}GAN-FE is available on GitHub.[1] This article contributes three major aspects that are listed as follows.

1) A novel convolutional two-stream WGAN-based hyperspectral feature extraction method is proposed. This method extracts latent embeddings by generating artificial spatial–spectral samples via two-stream WGAN. To the best of our knowledge, it is the first work that introduces this two-stream strategy into WGAN-based models for hyperspectral feature extraction.
2) Adapted from the classical two-stream ConvNet in the behavior recognition, a novel two-stream definition strategy is proposed, which is suitable for the hyperspectral data with a unique property. Our used loss function is also in accordance with this strategy.
3) The trained discriminator is transformed into the feature extractor without adding additional layers, reducing the computational complexity. Moreover, the latent dimension (LD) is changeable by adjusting the node number in the final layer.

[1]https://github.com/Geoscienceywb/sc2GAN-FE

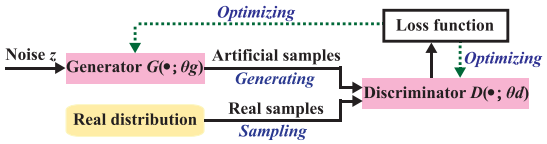


Fig. 1. Diagram of GAN, including one generator and one discriminator.

The remainder of this article is organized as follows. Section II introduces generative models and their improvements. Section III presents the methodology of cs^{2}GAN-FE. Section IV reports experimental results on publicly available hyperspectral data sets, followed by the conclusion in Section V.

II. Related Work and Motivation

Generative models generate samples randomly from the latent probability distribution in the target space. Techniques utilizing various generative models yield unusually brilliant results in pattern recognition domains. Typical generative models include variational autoencoders (VAEs) and GANs. VAE not only maintains the advantages of data reconstruction but also has the characteristics of generative models; however, it should ensure the similarity between inputs and outputs, and this requirement reduces its diversity. Compared with this stable model, GAN stands out in the aspect of generating latent discriminant features and shows great potential as well.

GAN [25] is regarded as one great invention in the deep learning domain (Fig. 1). It consists of two modules: one generator and one discriminator [7], [25]. The former generates artificial samples using low-dimensional noise z, and its internal structure is changeable. Generated samples share the same shape with real ones. The discriminator aims at distinguishing generated "fake" samples from real samples, and its inputs include two parts: artificial samples obtained by the generator and real samples in the raw space. In other words, the discriminator is a binary classifier. The generator tries to "cheat" the discriminator, making the latter believe that generated samples are real. In this adversarial environment, both of them could be trained and, finally, achieve a dynamic balance. The probability distribution of z is $p_z(z)$. The generator $G(\cdot;\theta_g)$ maps z into artificial samples whose probability distribution is $p_g(z)$. The inputs of the discriminator $D(\cdot;\theta_d)$ include two components: artificial samples [probability distribution is $p_g(z)$] and real samples x (probability distribution is p_{data}), where θ_g and θ_d are hyperparameters. The output of $D(\cdot;\theta_d)$ is a scalar; if it thinks that the input comes from the real data, the output is 1; if not, the output is 0. The loss function $L(D, G)$ [23] is shown as

$$\min_{\mathbf{G}}\max_{\mathbf{D}} L(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1-G(z))] \quad (1)$$

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The optimizing directions of D and G are contrary. To address the overfitting and instability when training GAN, Arjovsky and Bottou *et al.* [26] proposed WGAN by introducing the Wasserstein distance. Gulrajani *et al.* [27] improved it by introducing a gradient penalty, and its loss function is

$$\min_{\mathbf{G}} \max_{\mathbf{D}} L(D, G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[D(x)] - \mathbb{E}_{z\sim p_z(z)}[D(G(z))]. \tag{2}$$

Then, (2) is divided into loss functions for the discriminator and the generator as follows:

$$L(D) = \mathbb{E}_{z\sim p_z(z)}[D(G(z))] - \mathbb{E}_{x\sim p_{\text{data}}(x)}[D(x)] \tag{3}$$

$$L(G) = -\mathbb{E}_{z\sim p_z(z)}[D(G(z))]. \tag{4}$$

Gulrajani *et al.* [27] show that the prerequisite is that the discriminator is Lipschitz smooth in sample spaces [7], and a gradient penalty is added into (3) as

$$L(D) = \mathbb{E}_{z\sim p_z(z)}[D(G(z))] - \mathbb{E}_{x\sim p_{\text{data}}(x)}[D(x)] + k \times \mathbb{E}_{x\sim\Psi}[|\nabla D(x)|_l - m]^2 \tag{5}$$

where k is the weight of the gradient penalty, Ψ is the space of real samples, and $|\nabla D(x)|_l$ is the l-norm of the gradient of the discriminator. We hope that $|\nabla D(x)|_l$ is small enough.

As shown above, GAN has the ability to generate variable features and making them in accordance with real features by striking a balance between the generator and the discriminator. Considering this property of GAN, it is suitable for hyperspectral feature extraction. By generating deceptive samples for the discriminator, its discriminative ability can be raised through consecutive training. The trained discriminator can be transformed into the feature extractor with great classification capacity. Moreover, Tao *et al.* [23] used the ResNet mainstream and the auxiliary stream to extract hyperspectral features, providing the inspiration of our method. The above works are the motivation of this research.

III. Methodology

A novel convolutional two-stream WGAN-based hyperspectral feature extraction (cs^{2}GAN-FE) method was proposed. Each band corresponds to one wavelength, and the values are the spectral reflectance of pixels in the specific wavelength. These fixed locations of pixels are called static features. Similar to video data, the variance of the spectral reflectance in contiguous bands is continuous, and it reflects the dynamic property of pixels (Fig. 2). They are called dynamic features, reflecting the local changing trend of spectral reflection.

A. Convolution Layers for Two Streams

Convolution (Conv) layers extract spatial features by Conv kernels [6]. For these two streams, we chose different kernels.

1) The spatial organization of pixels in each band represents the correlation of target objects, and the spectral dimension reflects the continuity of wavelengths. Scholars tended to use 2-D/3-D kernels, and they both perform well. However, compared with the latter, 2-D kernels greatly promote the training efficiency, so they are selected for the static stream.

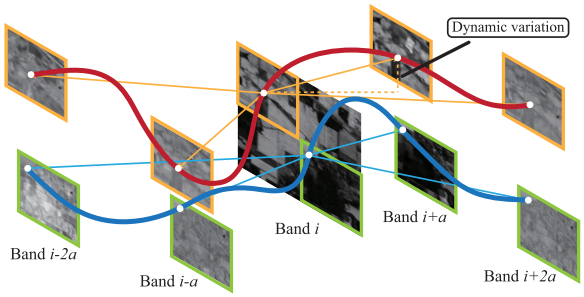


Fig. 2. Object reflectance variances transferred in the dynamic stream. The variances reflect the dynamic spectral reflectance changes of pixels rather than pixels themselves. Two variance lines (red and blue) in the nearest four bands are shown, and the difference of values is their dynamic variances.

2) Similar to video data in the behavior recognition domain (pixels are moving continuously), our spectral reflectance variance data contain abundant continuous features within adjacent bands.

Different from the spectral dimension in (1), the third dimension describes the local changing trend of spectral reflection, which should not be ignored. Thus, we selected 3-D kernels for the dynamic stream. These settings are also analyzed in Section IV.

B. Framework of cs^{2}GAN-FE

This whole structure (see Fig. 3) consists of two modules: one generator and one discriminator. The definitions of modules and streams are not separated: for each stream, half of it lies in the generator, and the other half lies in the discriminator. The analysis about kernels in Section III-A is applicable to all layers in both modules. The generator maps the low-dimensional noise into specific high-dimensional spaces and generates deceptive static/dynamic samples.

"Real" static/dynamic samples are obtained from real hyperspectral data (denoted as $\mathbf{X} \in \mathbb{R}^{a\times b\times c}$ comprising $a \times b$ samples and c bands); their shapes are $[a \times b, \zeta_n, \zeta_n, c]$ and $[a\times b, \zeta_n, \zeta_n, \iota, c]$, respectively. $a\times b$ is the number of samples, c is the number of spectral reflectance bands, and ζ_n is the size of the neighboring window (a $\zeta_n \times \zeta_n$ sliding window is used to select spatial neighbors). We use ι nearest bands ($\iota/2$ bands in front of it and $\iota/2$ bands behind it) to simulate the dynamic object reflectance variances. The discriminator aims to correctly distinguish "real" samples from "fake" samples.

C. Structure of the Discriminator in cs^{2}GAN-FE

The discriminator is shown on the right-hand side of Fig. 3. For the static stream part, the input shape is [batchsize, ζ_n, ζ_n, c] for each iteration. The batch size is the number of samples selected in $\{2^6, 2^7, \ldots, 2^{10}\}$ for training in each iteration. There are two 2-D convolution layers. The first one consists of 256 2-D 5×5 kernels (the stride is $[1, 1]$). The second layer includes 2-D 5×5 kernels, their number equals to the expected dimension of latent features, and the stride is $[1, 1]$, so the output shape is [batchsize, 1, 1, LD]. For the dynamic stream part, the input shape is [batchsize, $\zeta_n, \zeta_n, \iota, c$], and there are two 3-D convolution layers. They both consist of 3-D $5 \times 5 \times 2$ kernels, and

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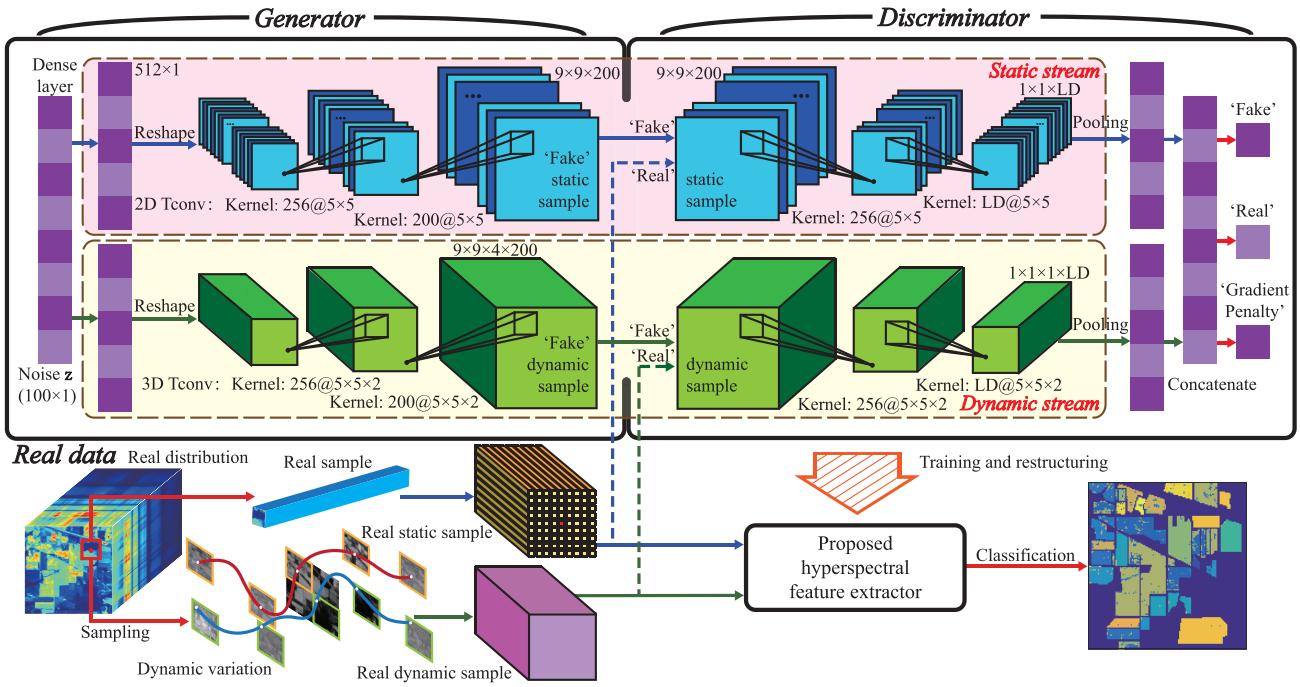


Fig. 3. Framework of the proposed cs^{2}GAN-FE (taking the Indian pines data set for example, $c = 200$, $\zeta_n = 9$, and $\iota = 4$). The left module is the generator, the right one is the discriminator, and two streams are boxed by dotted lines. Convolution kernels are represented as kernel number@kernel shape.

their numbers are 256 and LD, respectively. Outputs of two streams are flattened and then concatenated in a feasible way. "Fake" and "real" samples are inputted, and this module tries to distinguish them correctly. Layer normalization is chosen, and the Leaky rectified linear unit (leakyRelu) is selected as the activation function in this module [7].

D. Structure of the Generator in cs^{2}GAN-FE

The generator is shown on the left-hand side of Fig. 3. It maps the low-dimensional noise into high-dimensional spaces and generates deceptive samples. It guarantees the diversity of samples and guides the training direction. The noise is a random vector whose elements are within $[-1, 1]$. Two fully connected layers with 512 nodes generate initial inputs for two streams, respectively. This module is similar to the discriminator, and the whole network is symmetric (similar to traditional AEs). Transposed Conv (Tconv) layers are used to enlarge the size of inputs, so they are upsampling layers actually. Two Tconv layers in the static stream part have 256 and c 2-D 5×5 kernels, respectively, where c is the number of bands, which is changeable when using various data sets. In the dynamic stream, layers have 256 3-D kernels ($5\times5\times2$) and c 3-D kernels ($5\times5\times2$), respectively. We use 2-D 5×5 kernels and 3-D $5\times5\times2$ kernels since we assume $\zeta_n = 9$ in Fig. 3.

E. Structure of the Proposed Feature Extractor

After training the whole network, we remove the generator, and then, the discriminator becomes a practical feature extractor. Their differentiation lies in the uselessness of "fake" samples in the latter one. "Real" samples are inputted into the feature extractor, and static/dynamic features are extracted through multiple layers. Then, the concatenation layer merges multiple extracted features to obtain the latent embeddings for the following classification.

F. Loss Function in cs^{2}GAN-FE

Considering the designed two-stream strategy, we modified the loss functions in (4) and (5) as $L(G) = -\mathbb{E}_{z\sim p_z(z)}[D(G(z))]$ for the generator and

$$\begin{aligned} L(D) &= \mathbb{E}_{z\sim p_z(z)}[D(G(z))] - \mathbb{E}_{x\sim p_{\text{data}}(x)}[D(x)] \\ &\quad + k \times \mathbb{E}_{x\sim\Psi}[|\nabla D_{\text{static}}(x)|_l + |\nabla D_{\text{dynamic}}(x)|_l - 2 \times m]^2 \end{aligned} \tag{6}$$

for the discriminator, where letters share the same meanings with traditional ones, and we set $k = 10$ and $m = 1$ in this article. Notably, there are two streams in this architecture, so we use $\nabla D_{\text{static}}(x)$ and $\nabla D_{\text{dynamic}}(x)$ to simulate their gradients.

$L(G)$ here has the same definition with (4), which is the negative expectation $(-\mathbb{E}(\cdot))$ of the discriminator output $D(G(z)$ (the discriminator is fixed now). In (5), $L(D)$ consists of two parts: $\mathbb{E}_{z\sim p_z(z)}[D(G(z))] - \mathbb{E}_{x\sim p_{\text{data}}(x)}[D(x)]$ and the gradient penalty. The former is calculated with the expectation of the discriminator outputs using both "fake" and "real" samples, so this part is not changed since our network also has one output. The latter reflects the gradient from each input to the output, and the discriminator in cs^{2}GAN-FE contains two inputs, so two gradients are provided as $\nabla D_{\text{static}}(x)$ and $\nabla D_{\text{dynamic}}(x)$, respectively. They are calculated by Keras.gradients() with the Keras framework, and they should be smaller than m.

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G. Training Process on the Proposed Network

Although elements in the whole architecture are fully connected, the generator and the discriminator are not trained together but in an alternating way. First, we use the generator with initial parameters to generate enough deceptive (or "fake") static/dynamic samples and then use them to train the discriminator. Notably, the generator will not be considered when training the discriminator. After that, we fix the discriminator with trained parameters to train the generator. They are trained successively in multiple iterations. Raw samples (normalized into $[-1, 1]$) are randomly separated into small batches, and all generated batches are utilized for the whole training process. The ADAM is used, and the learning rate is set as 5×10^{-4}.

The computational complexity of cs^{2}GAN-FE is $\mathcal{O}(\zeta_n^4 \iota^2 c)$, where ζ_n is the sliding window size, ι is the number of nearest bands, and c is the band number of hyperspectral data sets. It shows that the training time will be easily influenced by ζ_n, so the selection of sliding windows is quite important for cs^{2}GAN-FE. The computing time will also be affected by other factors (such as the convergence of loss functions, the number of epochs, and the data set types), so the training time is hard to be predicted just by the computational complexity before related experiments. The training time per iteration is shown to evaluate the training efficiency in the following experiments.

IV. Experimental Results and Analysis

A. Experimental Settings

Four publicly available hyperspectral data sets are chosen,[2,3] [28], [29] and all experiments are carried out using Python 3.7.0 (Spyder) on a PC with Intel Core i7-6700 CPU and 8-GB RAM. Some state-of-the-art techniques are chosen for benchmarking, including AE, VAE, GAN, WGAN, spatial GAN (s-GAN), spatial WGAN (s-WGAN), DCGAN [22], [30], and uWGAN [7]. AE and VAE both have symmetric structures, and three fully connected layers exist in each module (tanh for activation). GAN and WGAN are similar with cs^{2}GAN-FE with the same corresponding parameters. For s-GAN and s-WGAN, only the static stream exists to extract spectral–spatial features. DCGAN and uWGAN have almost the same structures with [7] and [22], respectively. DCGAN uses the same feature extractor as uWGAN.

For all benchmarks, the best classification results (generally within 10^4 iterations) are recorded for the sake of fairness. 30% samples are randomly selected for training, and all samples are used for testing for completeness. SVM is applied for classification using LIBSVM,[4] and fivefold cross validation is utilized for optimal parameter selection. The parameters (penalty coefficient and gamma in RBF kernels) are selected from 2^{-4} to 2^4. 10% samples in each class are randomly selected for training (the remaining 90% are for testing). It means that the whole samples are divided into ten parts, and they are selected in turn for training to repeat the experiments ten times (to reduce the influence of training sample selection). Overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are calculated for detailed comparison.

[2]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[3]http://dase.grss-ieee.org/

[4]https://www.csie.ntu.edu.tw/ cjlin/libsvm/

TABLE I

Classification Accuracies (%) With Different ζ_n's. Bold Results Are the Best

ζ_n	3	5	7	9	25
OA	77.36±0.06	82.89±0.02	85.83±0.17	84.66±0.17	**94.83±0.11**
AA	65.20±0.15	68.66±0.01	73.73±0.23	75.33±0.32	**84.83±0.07**
κ	73.99±0.07	80.40±0.02	83.77±0.20	82.44±0.20	**94.11±0.12**

TABLE II

Classification Accuracies (%) With Different ι's. Bold Results Are the Best

ι	2	4	6	8
OA	83.34±0.07	84.66±0.17	85.52±0.29	**89.85±0.19**
AA	70.51±0.34	75.33±0.32	73.89±0.20	**79.07±0.003**
κ	80.91±0.09	82.44±0.20	83.40±0.34	**88.39±0.22**

B. Hyperparameter Analysis

1) Window Size Analysis: The hyperparameter (the sliding window size ζ_n, the number of nearest bands ι, and the training proportion between two modules) selection is crucial to the spatial feature exploitation, and the Indian Pines data is chosen for analysis. To show the effect of ζ_n on the result, we set ι as 4, the proportion is 1:1, and ζ_n is chosen in $[3, 5, 7, 9, 25]$ ($\zeta_n = 25$ shows the result with a big sliding window [7], [11], [12]). Accuracies are calculated using features with different ζ_n's (see Table I). Higher accuracies and lower standard deviations correspond to better results. It shows that the accuracies rise with the increase of ζ_n since more spatial features are considered. However, the training time per iteration is about 120 s when $\zeta_n = 25$, which is much longer than that when $\zeta_n = 9$ (about 2.5 s). It is clear that too large ζ_n's seriously impact the efficiency, so smaller ζ_n's are considered in this article. After carefully consideration, we set $\zeta_n = 9$.

2) Nearest Band Number Analysis: We set $\zeta_n = 9$, and the proportion is 1:1. ι is chosen in $[2, 4, 6, 8]$. Experiments show that the results become better when ι increases from 2 to 4. When $\iota = 4$ and $\iota = 6$, similar results are obtained. Although all accuracies reach the peak when $\iota = 8$, using too many nearest bands will make the whole process time-consuming (see Table II). It shows that, when using two, four, and six bands, the training time for each iteration remains stable; however, more than 3 s are needed when eight bands are considered. Finally, we set $\iota = 4$.

3) Training Proportion Analysis: The training proportion between the generator and the discriminator is also crucial. We set $\zeta_n = 9$ and $\iota = 4$ and change the training proportion (1:1, 5:1, and 1:5). Results are shown in Table III. 5:1 means that the generator is trained five times, and then, the discriminator is trained one time in each iteration. When the generator is trained more times than the discriminator, the results are not satisfactory (OA = 67.66%). When using 1:5, the OA is 3.33% higher than 1:1, but AA is 1.38% lower. Moreover, using 1:5 needs more training time, so, finally, we choose 1:1.

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TABLE III

CLASSIFICATION ACCURACIES (%) WITH DIFFERENT TRAINING PROPORTIONS BETWEEN THE GENERATOR AND THE DISCRIMINATOR. BOLD RESULTS ARE THE BEST

proportion	1:1	5:1	1:5
OA	84.66±0.17	67.66±0.08	**87.99±0.02**
AA	**75.33±0.32**	51.98±0.01	73.95±0.14
κ	82.44±0.20	62.39±0.07	**86.24±0.03**

C. Experiments on the Indian Pines Data Set

In this section, hyperparameters are set as follows: $\zeta_n = 9$, $\iota = 4$, and the training proportion is 1:1. Classification results (see Table IV, LD = 30, and more choices are provided later) show that cs^{2}GAN-FE (cs^{2}GAN for short) raises the classification ability with the SVM classifier. OA reaches 84.66% by cs^{2}GAN-FE when LD = 30, which is 2.49% and 5.13% bigger than AE-based networks (AE and VAE). Although VAE generates new characteristics when training, AE extracts better latent features by its strict loss function. The performances of GAN and WGAN are similar although the former has higher accuracies. When using other data (see Sections IV-D and IV-E), WGAN sometimes works better than GAN. It is obvious that GAN is more suitable for the Indian Pines data set than WGAN; however, their features have lower accuracies because the spatial information is not extracted, showing the importance of utilizing the spatial distribution. DCGAN and uWGAN perform similar with cs^{2}GAN-FE, but the dimensions of their latent features (are fixed to 768 in [7]) are far bigger, and more time is needed for classification (SVM will need more time to construct hyperplanes). The training time per iteration is provided in Table V. Although using two streams needs more training time (about 2 s), the improved performances prove that this costing is tolerable and valuable. Performances of static/dynamic streams show that using both streams could relatively increase OA/AA/κ, and our two-stream strategy is feasible in this domain.

The widely used random train/test split procedure is chosen in our experiments. Audebert *et al.* [9], [31] proposed that this sampling may not be the most reliable choice in this domain. They found that this split tends to include testing samples in the receptive fields of training samples when choosing CNN-based networks, and they designed the disjointed train/test split strategy.[5] Using this split, spectral–spatial features are extracted only from disjointed parts of the whole data. The cs^{2}GAN result with the disjointed split (the subscript DJ means using the disjointed split) is provided in Table IV. Training samples in cs^{2}GAN$_{\mathrm{DJ}}$ are specifically chosen, and it is inevitable to omit some discriminant information. Except for DCGAN and uWGAN, it works better than other methods with the random split, so DCGAN$_{\mathrm{DJ}}$ and uWGAN$_{\mathrm{DJ}}$ are also shown (see Table IV and Fig. 4). cs^{2}GAN$_{\mathrm{DJ}}$ works better than them (it increases OA by 6.76% and 2.73%, respectively); results show that DCGAN meets with difficulties when using the disjointed split. However, cs^{2}GAN performs well because of the full utilization of spatial information.

The two-stream strategy is shown in Section III; we use 2-D kernels in the static stream and 3-D kernels in the dynamic stream. To show the importance of this two-stream strategy, we only use 2-D/3-D kernels in both two streams and compare their results with cs^{2}GAN-FE (see Fig. 5). We also directly use the normalized raw data to replace the proposed spectral reflectance variance in the dynamic (3-D) stream, showing the benefit of our strategy. To use different kernels, the input shapes are slightly changed for adaptation, such as the 3-D kernel-based dynamic stream input has a [batchsize, 9, 9, 4, 200] shape and its shape is changed into [batchsize, 9, 9, 4×200] to use the 2-D kernels. Fig. 5(a) has too much noise and too many wrongly classified samples since the dynamic features are not fully extracted. Fig. 5(c) uses raw samples as inputs for both streams, and it performs worse

[5]https://github.com/nshaud/DeepHyperX

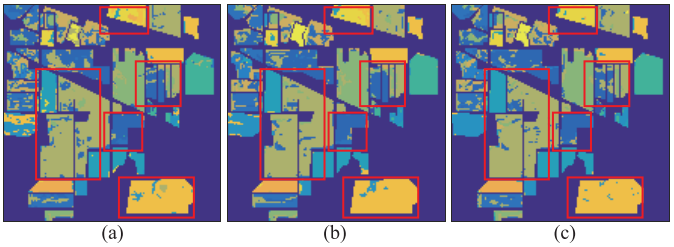


Fig. 4. Classification maps using (a) DCGAN$_{\mathrm{DJ}}$, (b) uWGAN$_{\mathrm{DJ}}$, and (c) cs^{2}GAN$_{\mathrm{DJ}}$. Five typical parts are marked by red squares.

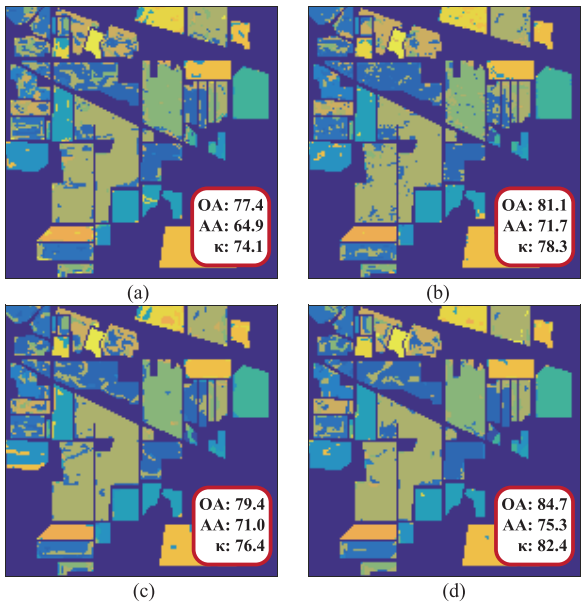


Fig. 5. Classification maps and accuracies using (a) two streams with 2-D kernels, (b) two streams with 3-D kernels, (c) two streams with 2-D kernels and 3-D kernels (their inputs are both raw samples), and (d) our proposed streams with 2-D kernels and 3-D kernels in cs^{2}GAN-FE, respectively.

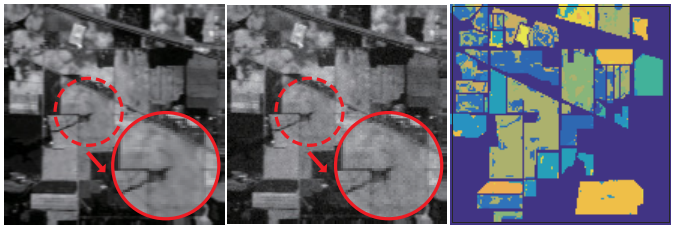


Fig. 6. Classification performance using the Indian pines data set with the random white Gaussian noise, including (a) and (b) 30th band with noise and (c) classification map by cs^{2}GAN-FE. One typical part (encircled) is enlarged to show the noise intensity.

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TABLE IV

CLASSIFICATION ACCURACIES (%) BY COMPARISON METHODS FOR THE INDIAN PINES DATA SET (LD = 30). BOLD VALUES INDICATE THE BEST RESULTS

1	AE	VAE	GAN	WGAN	sgan	swgan	dcgan$_{DJ}$	dcgan	uwgan$_{DJ}$	uwgan	static	dynamic	cs^{2}GAN$_{DJ}$	cs^{2}GAN
1	17.49	21.63	10.91	2.89	27.88	42.60	39.98	42.19	22.05	38.51	22.64	21.59	**49.19**	43.23
2	78.03	73.76	69.59	61.61	67.84	68.04	67.66	77.53	72.48	77.32	72.73	66.72	73.38	**79.58**
3	67.79	64.50	53.90	44.10	53.16	51.80	60.08	**74.19**	65.65	69.68	61.83	62.40	65.99	67.45
4	62.06	56.03	43.21	40.12	32.38	42.69	43.83	71.80	46.58	66.59	51.86	52.27	53.47	**68.71**
5	88.44	87.57	81.38	79.50	73.46	72.26	56.87	77.91	76.97	**88.82**	76.43	57.34	87.15	86.00
6	94.90	94.07	91.86	89.45	91.00	92.44	90.85	92.88	93.07	92.24	90.03	90.90	94.66	**95.48**
7	**36.52**	32.61	9.57	10.87	10.00	9.57	17.39	18.26	11.74	17.39	12.17	1.74	23.48	12.61
8	**99.07**	98.77	97.95	97.86	96.61	95.66	92.34	98.14	97.07	97.30	96.71	93.46	96.25	96.93
9	56.25	**70.63**	16.88	0.63	34.38	41.88	14.37	16.25	15.63	37.50	36.25	11.87	21.88	40.63
10	76.05	70.22	59.21	68.09	71.11	61.69	74.89	**79.64**	79.02	78.73	74.54	63.20	74.48	77.50
11	85.10	81.27	79.50	81.24	83.62	80.12	86.44	90.19	86.58	89.96	87.88	82.75	90.23	**90.40**
12	**72.46**	69.31	62.51	43.82	49.73	45.91	52.53	68.54	52.13	55.35	44.86	56.31	55.10	64.20
13	93.37	93.84	95.16	95.58	86.95	94.37	89.37	98.11	95.00	92.05	94.95	88.79	95.11	**98.74**
14	94.66	97.39	96.71	96.07	92.09	93.66	90.61	96.15	93.12	**98.13**	92.45	95.58	95.52	96.57
15	62.08	55.00	44.59	35.50	57.66	66.84	60.61	67.84	67.40	89.12	59.47	83.45	**89.85**	87.25
16	88.82	86.71	85.06	80.82	91.29	33.41	41.41	96.47	**100.00**	68.82	45.29	98.82	92.82	**100.00**
OA	82.17	79.53	74.80	72.19	74.99	73.45	75.53	84.03	79.56	83.70	77.77	75.94	82.29	**84.66**
AA	73.32	72.08	62.37	58.01	63.70	62.06	61.20	72.88	67.16	72.34	63.76	64.20	72.41	**75.33**
κ	79.59	76.52	70.98	67.93	71.28	69.48	71.82	81.74	76.61	81.32	74.42	72.33	79.67	**82.44**

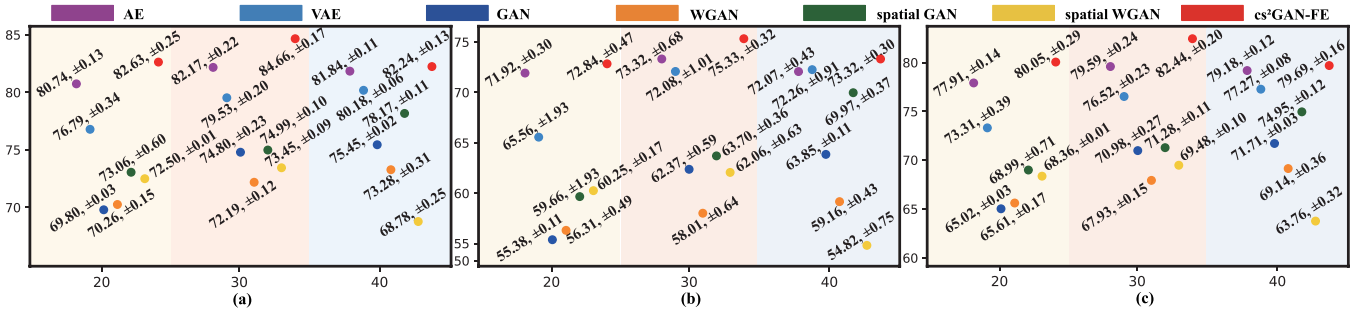


Fig. 7. Classification accuracies of the Indian Pines data set obtained by comparison methods versus the latent dimension. (a) Overall accuracy (%). (b) Average accuracy (%). (c) Kappa coefficient (%).

TABLE V

TRAINING TIME (S) PER ITERATION FOR COMPARISON METHODS (BATCH SIZE IS 512)

Methods	Time per iteration	Methods	Time per iteration
AE	0.00033	DCGAN	0.36524
VAE	0.02822	uWGAN	0.48540
GAN	0.04102	cs^{2}GAN-FE (2)	2.20974
WGAN	0.04690	cs^{2}GAN-FE (4)	2.49733
s-GAN	0.22750	cs^{2}GAN-FE (6)	2.58102
s-WGAN	0.44497	cs^{2}GAN-FE (8)	3.04618

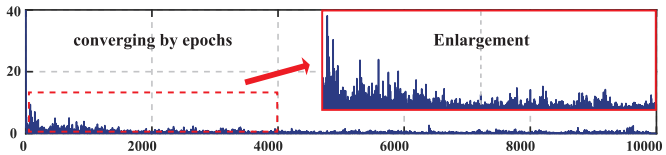


Fig. 8. Loss function of the discriminator versus the number of epochs.

than cs^{2}GAN-FE [see Fig. 5(d)], which chooses the spectral reflectance variance as the dynamic input. Compared with using raw data directly, our input clearly reflects the dynamic changing rule in adjacent bands, making the extracted dynamic features contain more diverse information than the static features. cs^{2}GAN-FE Fig. 5(d) works better than Fig. 5(b) although the result in Fig. 5(b) is also great; however, using 3-D kernels in both streams needs too much time. The training times per iteration in Fig. 5(a)–(d) are 2.03, 3.60, 2.55, and 2.50 s, respectively. This shows that the kernels used in our strategy are efficient.

The random white Gaussian noise with a small signal/noise radio (SNR = 30 dB) is added into raw data to show the antinoise performance of cs^{2}GAN-FE when extracting features from hyperspectral data with noise (its intensity is shown in Fig. 6). Compared with Fig. 6(a), Fig. 6(b) contains too much noise, disturbing the detection of samples. The added noise provides unreliable and deceptive spectral–spatial information; however, cs^{2}GAN-FE performs well (OA: 77.07%, AA: 64.15%, and κ: 73.66%), and Fig. 6(c) shows its strong antinoise ability.

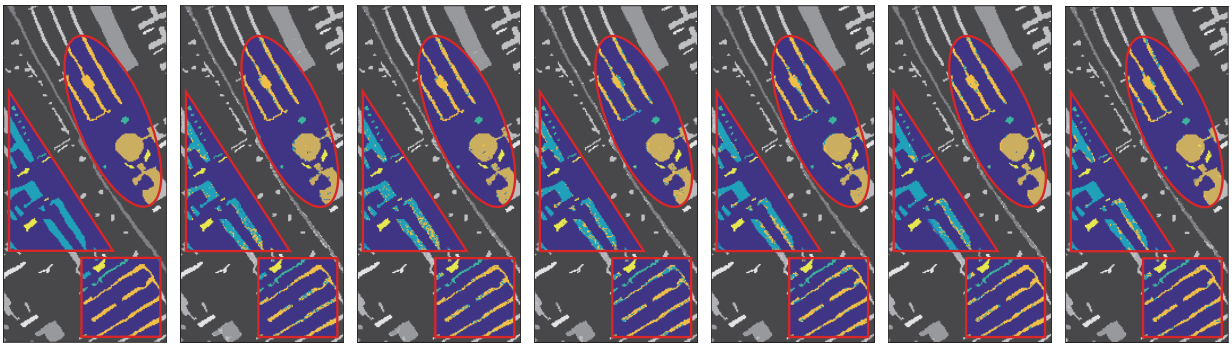
The variations of classification accuracies with respect to the LD are shown in Fig. 7. LD varies from 20 to 40 to reduce the effect of LD selection. DCGAN and uWGAN are not included since their LDs are fixed to 768. Points in each square (light yellow/red/blue) share the same LD, and they are spread out in sequence to show them clearly since the point distribution is quite narrow. The x-axes of these points have no meanings, and their y-axes have values, which are also marked near them. Figures show that cs^{2}GAN-FE works well with multiple LDs, and its performance is stable. AE generates great results with relatively higher accuracies, but it is more volatile than cs^{2}GAN-FE

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TABLE VI

CLASSIFICATION ACCURACIES (%) BY COMPARISON METHODS FOR THE PAVIA UNIVERSITY DATA SET (LD = 30). BOLD VALUES INDICATE THE BEST RESULTS. (5N) MEANS THAT THE RANDOM WHITE GAUSSIAN NOISE IS ADDED INTO THIS DATA SET

	Raw	AE	VAE	GAN	WGAN	s-gan(5)	s-wgan(5)	static(5)	dynamic(5)	cs^{2}GAN-FE			
										(3)	(5)	(5N)	(7)
1	69.37	74.73	76.88	71.82	73.05	67.84	79.33	73.81	74.89	**83.31**	81.47	76.08	82.85
2	97.93	99.48	99.31	96.98	98.10	98.88	99.48	96.63	98.70	98.96	99.83	98.79	**99.74**
3	78.42	80.50	81.91	80.68	76.41	77.26	79.58	79.46	74.63	84.96	84.17	80.93	**89.00**
4	99.35	98.48	98.48	98.48	98.92	98.70	**99.57**	94.58	97.40	98.70	98.05	97.25	96.10
5	99.08	99.08	99.08	99.39	99.39	95.40	97.85	**100.00**	99.69	99.69	**100.00**	99.85	**100.00**
6	92.77	94.92	93.45	90.71	92.38	88.86	89.44	90.13	**95.60**	95.41	94.13	92.35	94.62
7	80.95	76.28	84.27	82.61	83.31	71.55	80.18	76.47	75.70	**84.72**	81.59	79.15	84.21
8	98.42	99.80	**100.00**	99.60	99.60	97.62	**100.00**	94.65	97.43	98.22	97.03	94.53	96.83
OA	86.86	87.39	89.38	87.54	87.28	83.80	87.69	85.44	85.74	90.87	89.71	87.34	**91.39**
AA	89.54	90.41	91.67	90.03	90.14	87.01	90.68	88.22	89.26	**93.00**	92.03	89.87	92.92
κ	84.35	84.98	87.36	85.17	84.87	80.71	85.35	82.68	83.05	89.14	87.77	84.93	**89.75**



(a) (b) (c) (d) (e) (f) (g)

Fig. 9. Classification maps of the Pavia University data set obtained by comparison methods. (a) Ground truth. (b) GAN. (c) WGAN. (d) Spatial GAN. (e) Spatial WGAN. (f) cs^{2}GAN(5). (g) cs^{2}GAN(5N).

TABLE VII

CLASSIFICATION ACCURACIES (%) BY COMPARISON METHODS FOR THE SALINAS DATA SET (LD = 30). BOLD VALUES INDICATE THE BEST RESULTS

	Raw	VAE	GAN	WGAN	s-gan	s-wgan	cs^{2}GAN
OA	92.63	93.06	91.88	92.03	89.08	92.37	**93.08**
AA	95.96	96.15	95.35	95.70	92.67	95.61	**96.23**
κ	91.79	92.26	90.95	91.12	87.82	91.49	**92.28**

TABLE VIII

CLASSIFICATION ACCURACIES (%) BY COMPARISON METHODS FOR THE DFC2018 DATA SET (LD = 20). BOLD VALUES INDICATE THE BEST RESULTS

	Raw	VAE	GAN	WGAN	s-gan	s-wgan	cs^{2}GAN
OA	80.23	83.32	86.12	86.23	84.90	84.44	**86.45**
AA	66.28	69.82	73.52	73.31	73.93	73.05	**74.00**
κ	76.36	80.03	83.43	83.59	81.99	81.43	**83.84**

(its OA increases by 4.69% when LD increasing from 20 to 30). The loss function of the discriminator when LD = 20 is shown in Fig. 8 with the first 10 000 epochs. It shows that the loss value decreases quickly before 2000 epochs, and then, the line converges to 0. This also illustrates that cs^{2}GAN-FE is easy to be trained using hyperspectral data.

D. Experiments on the Pavia University Data Set

The Pavia university data set (more complex scenarios than the first data set) is used, and the classification accuracies (LD = 30) are shown in Table VI. To remove the effect of ζ_n, we select it in $[3, 5, 7]$ and set $\zeta_n = 5$ in spatial GAN (s-gan) and spatial WGAN (s-wgan) for simplification. Moreover, the accuracies for the static line and dynamic line ($\zeta_n = 5$) are also shown to prove our feasibility. cs^{2}GAN-FE obtains better results (with higher OA/AA/κ) compared with other methods. When $\zeta_n = 7$, it raises OA/AA/κ compared with VAE by 2.01%/1.25%/2.39%. Samples in Label 1/2/3/5/7 are easy to be classified correctly by cs^{2}GAN-FE considering the detailed accuracies in multiple classes. When $\zeta_n = 5$, the best OAs of the static/dynamic features are 85.44% and 85.74%, which are both less than using our two-stream strategy.

Three main regions in the classification maps are lit by red lines in Fig. 9. cs^{2}GAN-FE performs almost the best in the triangle region (top area), and almost all methods have wrongly classified samples in its bottom area; however, spatial methods have relatively discrete noise compared with spectral methods. Samples in the yellow circular area in the bottom of the oval region are almost classified correctly by cs^{2}GAN-FE.

The random noise is also added: SNR = 30 dB and $\zeta_n = 5$; its accuracies and map are shown in Table VI (5N) and Fig. 9(g), respectively. The noise hinders the feature extraction process; however, better results are also generated by cs^{2}GAN-FE than some comparison techniques, such as WGAN and s-GAN. All results show our feasibility when processing noisy data.

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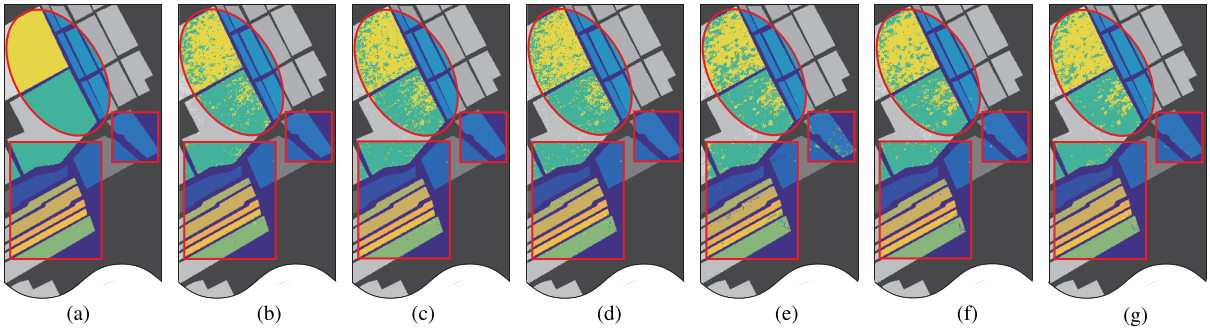


Fig. 10. Classification maps of the Salinas data set obtained by comparison methods; their bottom parts are hidden because of great accuracies. (a) Ground truth. (b) VAE. (c) GAN. (d) WGAN. (e) Spatial GAN. (f) Spatial WGAN. (g) cs^{2}GAN-FE.

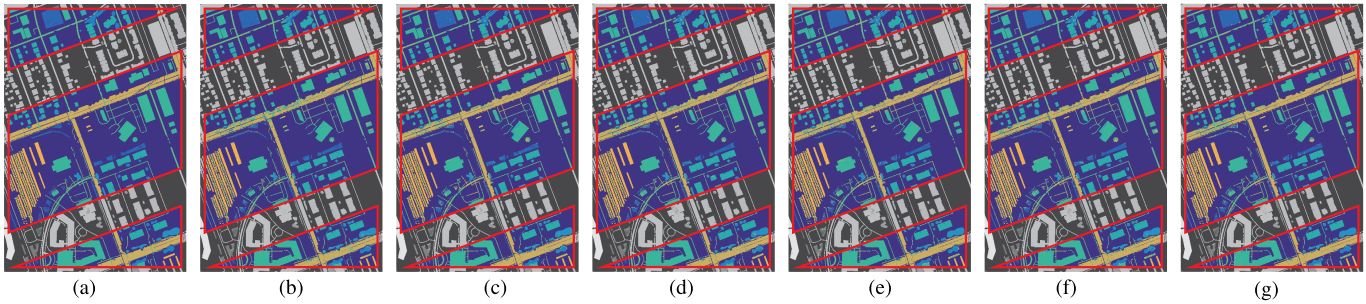


Fig. 11. Classification maps (partial) of the DFC2018 data set obtained by comparison methods. (a) Ground truth. (b) VAE. (c) GAN. (d) WGAN. (e) Spatial GAN. (f) Spatial WGAN. (g) cs^{2}GAN-FE.

E. Experiments on the Salinas Data Set

For this data set, we set $\zeta_n = 3$ without changing other hyperparameters. As shown in Table VII, cs^{2}GAN-FE is superior to others in terms of OA/AA/κ. VAE is a strong competitor of cs^{2}GAN-FE, and the latter one raises OA/AA/κ by 0.02%/0.08%/0.02%. Samples in most classes could be correctly classified by cs^{2}GAN-FE, showing its superiority. The classification maps are provided in Fig. 10. Although their differences are quite small, the noise of cs^{2}GAN-FE in the red oval, and the big square is relatively smaller than others. Moreover, more wrongly classified samples exist in the small square area by spatial GAN compared with our method.

F. Experiments on the DFC2018 Data Set

A part of the 2018 Data Fusion Contest (DFC2018) Data Set is used (LD = 20, $\zeta_n = 3$, and other parameters are not changed), and the results are shown in Table VIII. cs^{2}GAN performs better than other techniques in terms of classification accuracies. Three typical regions are lit (from top to bottom) in Fig. 11. Most methods behave similarly in the middle region; however, cs^{2}GAN works better in the first region and the last region (less wrongly classified samples exist), verifying the above conclusions.

V. CONCLUSION

A novel two-stream strategy-based hyperspectral feature extraction network cs^{2}GAN-FE is proposed. Different from a simple extension of the one-stream network, our two streams (static-dynamic streams) are carefully designed for various features: the former delivers static features in each band, and the latter uses the dynamic spectral reflectance variances of pixels within adjacent bands. Both features are effectively integrated, and the combined features are simulated comprehensively, working well for the following classification. cs^{2}GAN-FE is designed based on the WGAN, consisting of two modules: one generator and one discriminator. The former generates deceptive samples, while the latter distinguishes real samples from deceptive ones. The hyperparameters are carefully selected through quantitative analysis. Experiments are conducted on four hyperspectral data sets, and eight state-of-the-art techniques are chosen. cs^{2}GAN-FE works better using both the random and disjointed splits with respect to the accuracies and maps. It shows that cs^{2}GAN-FE has great potential in more hyperspectral tasks.

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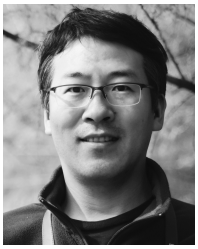
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